

Vector Subspace

Let V be an F -Vector space. A nonempty subset $W \subseteq V$ is called a vector subspace of V if W is also a vector space under the same addition and scalar multiplication. (Sometimes we may simply say subspace)

* Henceforth, we shall write αv for the scalar multiplication $\alpha * v$

Examples ① V itself is a subspace of the vector space V .

② $\{0\}$ is a subspace of any vector space V .

③ Let $\begin{pmatrix} 0 \\ 0 \end{pmatrix} \neq x \in \mathbb{R}^2$ be a point. Then the line

$L = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \alpha x : \alpha \in \mathbb{R} \right\} = \left\{ \alpha x : \alpha \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^2$.
 (Joining $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and x)

Proof: ① $\alpha_1 x, \alpha_2 x \in L$ then $\alpha_1 x + \alpha_2 x = (\alpha_1 + \alpha_2)x$, which again belongs to L

② Since associativity property holds in V , it also holds in L ($L \subseteq \mathbb{R}^2$)

③ $0 = 0 \cdot x \in L$, So $0 \in L$. (inverse $-(\alpha x) = (-\alpha)x$)

④ clearly for any $\alpha x \in L$, we have $(-\alpha)x \in L$ and $\alpha x + (-\alpha)x = 0$.

L is an abelian
subgroup of $\mathbb{R}^2$

Therefore, L is an abelian group. For any $\beta \in F$ and $\alpha X \in L$, we have $\beta(\alpha X) = (\beta\alpha)X \in L$. Thus L also has the same scalar multiplication.

* Note that $(V1)$, $(V2)$, $(V3)$ and $(V4)$ are automatically satisfied by L as they already hold for $\mathbb{R}^2$. $[as L \subseteq V]$

An observation

If $V \neq \{0\}$, that is, V has a nonzero element then V has infinitely many elements.

Proof

Let $0 \neq v \in V$. For any $\alpha, \beta \in F$ ($= \mathbb{R}$ or $\mathbb{Q}$ or $\mathbb{C}$) with $\alpha \neq \beta$, we note that $\alpha v \neq \beta v$ (why?).

Now since $\mathbb{R}$ has infinitely many elements, the set $\{\alpha v : \alpha \in F\}$ is also an infinite set.

[Let $\gamma := \beta - \alpha$ and observe that $\gamma \neq 0$

Then $\beta v = \alpha v \Rightarrow \gamma v = 0$. Since $\gamma \neq 0$, we must have $\gamma^{-1} \gamma v = \gamma^{-1} 0$, which implies $v = 0$, a contradiction.]

Definitions

Let V be an F -vector space

Linear Combinations:

Let S be a nonempty subset of V . An element $v \in V$ is said to be a linear combination of elements of S if

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n,$$

where $\alpha_1, \alpha_2, \dots, \alpha_n \in F$ and $v_1, v_2, \dots, v_n \in S$.

Span of a Set S .

Let S be a nonempty subset of V .

$$\text{Span of } S = \text{Span}(S) = \left\{ \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n : \begin{array}{l} \alpha_1, \alpha_2, \dots, \alpha_n \in F \\ v_1, v_2, \dots, v_n \in S \end{array} \right\}$$

$\text{Span}(S)$ is the set of all linear combinations of all elements of S .

Note: S may have infinitely many elements. However linear combinations involve any but only finitely many elements of S .

* Note that $\text{Span}(S) \subseteq V$.

Examples: ① Let $S = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \subseteq \mathbb{R}^2$.

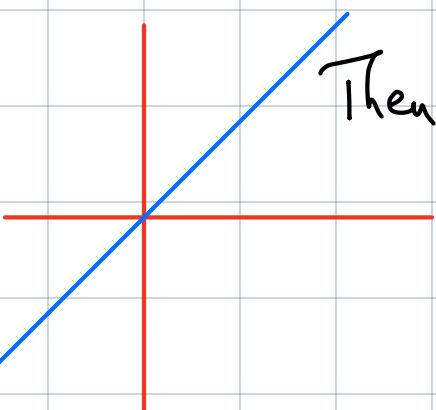
What is $\text{Span}(S)$?

Solution: Given any element $v = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{R}^2$, we observe that

$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, which is a linear combination of elements of S .

So, $\text{Span}(S) = \mathbb{R}^2$.

② Let $S = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} \subseteq \mathbb{R}^2$.



Then $\text{Span}(S) = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}$

$$= \left\{ \begin{pmatrix} \alpha \\ \alpha \end{pmatrix} : \alpha \in \mathbb{R} \right\}.$$

Line joining $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Note:

As discussed in today's class.

A non empty subset W of a Vector Space is a Subspace if $x+y \in W$ whenever $x, y \in W$ and $\alpha x \in W$ whenever $x \in W$ and $\alpha \in F$.